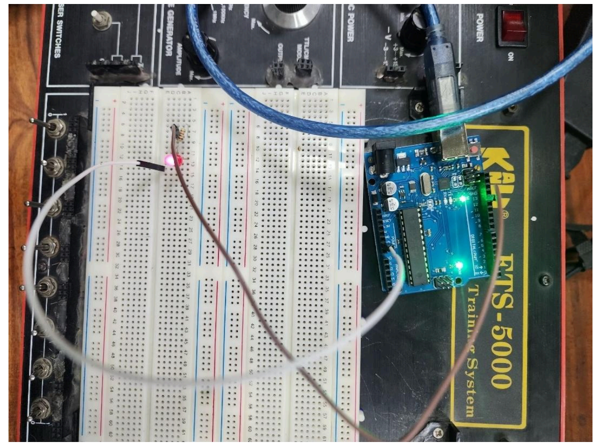
Lab 1

1) Write an Arduino program to turn ON an LED connected to pin 13.

```
void setup() {     // will run once
  pinMode(13, OUTPUT);
  digitalWrite(13, HIGH);
}

void loop() {    //infinte loop
  // Nothing to repeat
}
```



2) Write an Arduino program to blink an LED connected to pin 13 with a delay of 1 second.

```

void setup() {
  pinMode(13, OUTPUT);
}

void loop() {
  digitalWrite(13, HIGH); // LED ON
  delay(1000); // 1 second delay
  digitalWrite(13, LOW); // LED OFF
  delay(1000); // 1 second delay
}
```

3) Write an Arduino program to blink an LED 5 times with delay of 1 second each.

```
bool led = false;

void setup() {

  pinMode(13,OUTPUT);
}

void loop() {

 if(led==false){
      for(i=0;i<5;i++){
       digitalWrite(13,High);
       delay(1000);
       digitalWrite(13,High);
       delay(1000);
      }
      Led = true;
  }
}
```

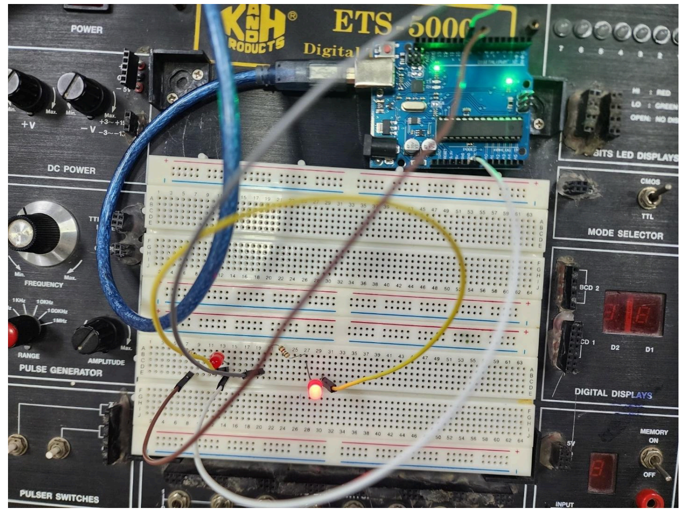
4) Write an Arduino program to blink two LEDs alternately connected to pins 8 and 9.

```

void setup() {
  pinMode(8, OUTPUT);
  pinMode(9, OUTPUT);
}

void loop() {
  digitalWrite(8, HIGH);
  digitalWrite(9, LOW);
  delay(1000);

  digitalWrite(8, LOW);
  digitalWrite(9, HIGH);
  delay(1000);
}
```



5) List down the items that you think they could be an IoT device.

An IoT device is any **physical** object that can **sense** data, **connect to the internet**, and **exchange** information.

Examples:

Smart light bulbs

Smart fans / ACs

Smart refrigerator

Smart washing machine

Smart TV

Smart door lock

Smart watch / fitness band

Smart electricity meter

Smart security camera

Smart irrigation system

Smart parking system

Smart smoke detector

6) Identify and analyze a device that is an IoT device now, but in the past was a non-IoT device. Describe and list the features of the device. Compare the functions of the device in the past to the functions of the device now.

Example: Electricity Meter

Past (Non-IoT Electricity Meter)

Features:

Manual meter reading

No internet connectivity

Paper-based billing

No real-time monitoring

Power theft detection not possible

Functions (Past):

Measures total energy consumption

Meter reader visits monthly

User gets bill after delay

Present (Smart IoT Electricity Meter)

Features:

Internet connectivity (Wi-Fi / GSM / NB-IoT)

Real-time power monitoring

Remote meter reading

Automatic billing

Power theft detection

Mobile app/web dashboard

Functions (Now):

Real-time energy usage tracking

Remote monitoring by utility company

Instant fault detection

Dynamic pricing support

User can monitor usage from phone

Comparison Summary:
Earlier meters only measured energy. Modern IoT meters measure, analyze, communicate, and optimize energy usage.

7) Describe the difference between Microprocessor and Microcontroller.

Feature	Microprocessor	Microcontroller
Definition	CPU only	CPU + RAM + ROM + I/O
External components	Required	Not required
Power consumption	High	Low
Cost	Expensive	Cheap
Size	Larger system	Compact
Usage	Computers,	laptops Embedded & IoT systems
Example	Intel Core i5	Arduino, PIC16, ESP32

8) What are the main characteristics of IOT?
Connectivity – Devices are connected to the internet

Sensing – Collects data using sensors

Data processing – Processes data locally or in cloud

Automation – Performs actions automatically

Real-time monitoring – Live data access

Scalability – Supports many devices

Intelligence – Uses AI/ML for decision making

9) Write the significance of Internet of things in society.
Improves quality of life

Saves time and energy

Enables smart cities

Improves healthcare monitoring

Enhances industrial automation

Reduces human effort

Helps in environment monitoring

Supports data-driven decisions

IoT makes systems smarter, faster, and more efficient.

10) What are the benefits and risks associated with Internet of things?

Benefits of IoT

Automation and convenience

Real-time monitoring

Energy efficiency

Cost reduction

Improved safety

Better decision making

Remote access and control

Risks of IoT

Security vulnerabilities (hacking)

Privacy issues

Data breaches

Dependence on internet

High initial cost

Compatibility issues

System failure risks